

NABARD Hackathon @ GFF 2026

TEAM NAME: c3codecoffeechaos

GRAMPULSE AI

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GramPulse AI

AI-Powered Rural Financial Intelligence Platform for Predictive Cash Flow, Risk Assessment & Decision Support

The Challenge

Rural micro-enterprises—including dairy farms, poultry units, handicraft businesses, village retail stores, and small agri-based enterprises—often face limited access to formal credit due to inadequate financial records and unpredictable income patterns.

Financial institutions rely primarily on historical data, making it difficult to identify emerging financial risks before they become critical. Additionally, external factors such as climate variability, seasonal demand, and fluctuating market prices significantly impact cash flow, yet these factors are rarely incorporated into existing financial assessment systems.

Limited Financial History

Rural enterprises lack the formal financial records required by conventional credit assessment systems.

Delayed Risk Detection

Financial distress is identified only after it occurs, leaving no room for proactive intervention.

Climate & Market Uncertainty

Income volatility driven by weather events and price fluctuations is rarely factored into assessments.

Inefficient Credit Processes

Existing credit assessment workflows are slow, static, and reactive rather than proactive.

GramPulse AI — Our Solution

GramPulse AI is an AI-powered Financial Decision Intelligence Platform that predicts future cash flow, assesses financial risks, and provides actionable recommendations for rural enterprises.

Unlike conventional systems that only evaluate past financial performance, GramPulse AI continuously analyzes financial, market, climate, and transaction data to generate intelligent forecasts and early warning alerts. The platform enables financial institutions, NABARD, and rural development agencies to make proactive, data-driven lending and intervention decisions.



Predict Future Cash Flow

30, 60, and 90-day forecasting for rural enterprises.



Detect Financial Stress Early

AI-driven early warning before loan default occurs.



Assess Credit Risk

AI-powered rural credit scoring and risk-adjusted planning.



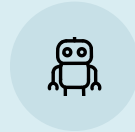
Climate & Market Analysis

Integrate environmental and market signals into financial models.



Recommend Interventions

Suggest suitable financial products and government schemes.



AI-Powered Decision Support

Unified intelligence core for institutions and field officers.

Why GramPulse AI?

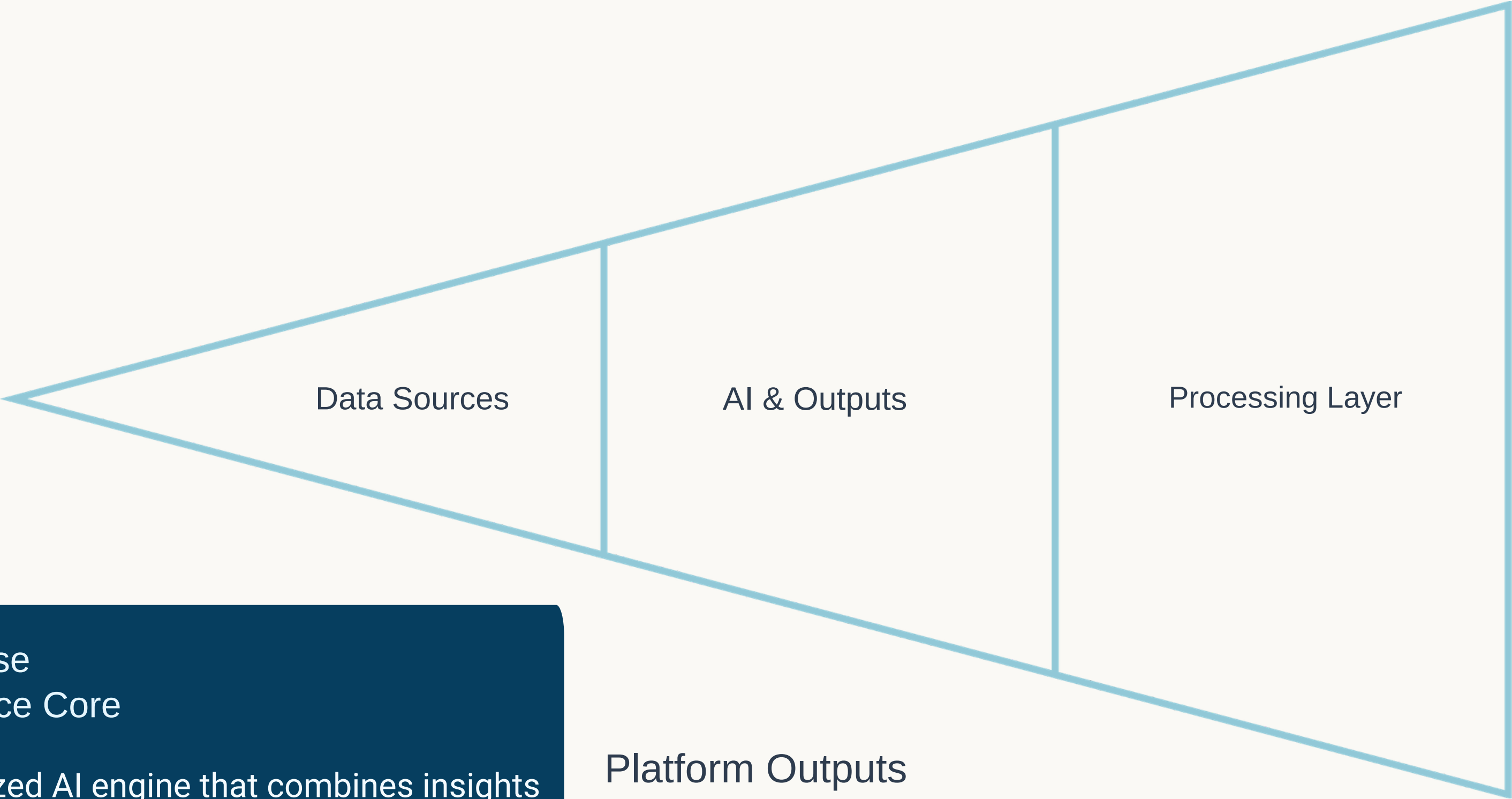
Existing Approach vs GramPulse AI

KEY INNOVATION

From Reactive Lending → Predictive Financial Intelligence

Traditional Systems	GramPulse AI
Uses historical financial records	Uses financial, climate, market & digital transaction data
Detects problems after they occur	Predicts financial stress before it occurs
Static financial analysis	Continuous AI-driven monitoring
Limited decision support	Intelligent recommendations & intervention planning
Separate analytical tools	Unified AI Decision Intelligence Platform

System Architecture — GramPulse AI Workflow



GramPulse
Intelligence Core

A centralized AI engine that combines insights from all specialized models to generate unified financial intelligence, recommendations, and decision support.

Platform Outputs

- AI Copilot for farmers and field officers
- Farmer Dashboard with personalized insights
- Field Officer Dashboard for portfolio monitoring
- Early Warning Alerts for at-risk enterprises
- Intervention Recommendations for institutions

Core Modules

1

Cash Flow Forecasting

Objective: Predict financial performance for the next **30, 60, and 90 days**.

- AI-based cash flow forecasting
- Income and expense prediction
- Scenario simulation
- Market trend forecasting
- Profit optimization recommendations

2

Risk Flagging

Objective: Identify enterprises at risk of financial stress before loan default occurs.

- Loan default prediction
- Financial stress scoring
- AI-powered rural credit score
- Smart loan recommendations
- Risk-adjusted repayment planning

3

Climate & Market Intelligence

Objective: Understand how environmental and market conditions influence financial stability.

- Weather analysis & rainfall monitoring
- Crop cycle tracking
- Market price intelligence
- Climate risk alerts

4	Explainable AI Objective: Provide transparent and trustworthy AI predictions. • SHAP-based explanations • Prediction confidence score • Risk factor visualization • What-if scenario analysis • Personalized financial recommendations
5	AI Copilot Objective: Deliver an intelligent assistant for farmers and field officers. • Multilingual chatbot • Voice interaction • Government scheme recommendations • Market advisory • Financial guidance
6	Decision Dashboard Objective: Provide real-time monitoring and decision support. • Village risk heatmap • Enterprise monitoring & priority ranking • Early warning alerts • Intervention recommendations

Technology Stack

Frontend

React.js · Streamlit · HTML ·
CSS · JavaScript

Backend

FastAPI · Flask

Database

PostgreSQL · MongoDB

Deployment

Docker · AWS · Google Cloud ·
Render

Machine Learning

- Python
- Scikit-learn
- XGBoost
- LightGBM
- LSTM

Explainable AI

SHAP — for transparent,
interpretable model outputs

Visualization

Plotly · Matplotlib

Business Model

Target Customers

- NABARD
- Regional Rural Banks
- Cooperative Banks
- Microfinance Institutions
- State Rural Development Agencies

Revenue Model

- Annual SaaS Subscription
Recurring platform access for financial institutions and agencies.
- Enterprise Licensing
Tailored licensing agreements for large-scale institutional deployments.
- Premium Analytics Services
Advanced reporting, custom model training, and deep-dive analytics.
- API Integration
Modular API access for embedding GramPulse intelligence into existing systems.
- Custom Dashboard Solutions
Bespoke dashboards designed for specific institutional workflows.

Expected Impact

Farmers & Rural Enterprises

- Improved access to formal credit
- Better financial planning
- Early financial risk detection
- Personalized recommendations

Financial Institutions

- Improved loan assessment
- Reduced default rates
- Better portfolio monitoring
- Data-driven lending decisions

Government & NABARD

- Early identification of vulnerable regions
- Efficient allocation of subsidies
- Better rural development planning
- Improved financial inclusion

Innovation & Competitive Advantage

What Makes GramPulse AI Unique?



Centralized AI Decision Intelligence Core

A unified engine combining all model outputs into a single, coherent intelligence layer.



Multi-Source Data Integration

Financial, climate, market, and digital transaction data fused into one platform.



Predictive Cash Flow Forecasting

30, 60, and 90-day forward-looking financial projections for rural enterprises.



AI-Driven Financial Risk Assessment

Proactive stress scoring and default prediction before problems escalate.



Explainable AI

SHAP-based transparency ensures every prediction is interpretable and trustworthy.



Multilingual AI Copilot

Voice and text assistant accessible to farmers across diverse linguistic regions.



GramPulse AI transforms rural finance from reactive decision-making to proactive financial intelligence.

Future Roadmap

Planned Enhancements



Our Vision

To build India's most intelligent rural financial decision-support platform that empowers farmers, financial institutions, and government agencies with predictive insights, transparent AI, and proactive interventions, fostering financial inclusion and sustainable rural development.

"Predict Early. Act Smarter. Empower
Rural Growth."

Thank You.